

Nutrient and methyl mercury exposure from consuming fish.

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There is controversy about the risks and benefits of consuming fish. Fish consumption provides nutrients, some of which are essential for brain growth and development. All fish, however, contain methyl mercury (MeHg), a known neurotoxicant. The toxic effect of MeHg seems most damaging during brain development, and thus, prenatal exposure is of greatest concern. At present the level of prenatal exposure associated with risk to a child's neurodevelopment is not known. Balancing the rewards and possible risks of fish consumption presents a dilemma to consumers and regulatory authorities. We review the nutrients in fish that are important in brain development and the current evidence of risk from MeHg at exposure levels achieved by consuming fish. We then review the findings from a large prospective cohort study of a population that consumes fish daily, the Seychelles Child Development Study. The MeHg content of the fish consumed in the Seychelles is similar to that of ocean fish available in industrialized countries, so they represent a sentinel population for any risk from fish consumption. In the Seychelles, evaluations of the children through 9 y of age show no consistent pattern of adverse associations with prenatal MeHg exposure. Recent studies in the Seychelles have focused on nutrients in fish that might influence a child's development, including long-chain polyunsaturated fatty acids, iodine, iron, and choline. Preliminary findings from this study suggest that the beneficial influence of nutrients from fish may counter any adverse effects of MeHg on the developing nervous system.